

F24-054-D-IlmPath

Project Team

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Chapter 1

Introduction

The purpose of this document is to provide an overview of our project, named IlmPath, discussing how we intend to complete it. The document will share insight on the problem which became a factor in coming up with the idea and developing on it. Furthermore the document will shine light on the scope as well the modules that the project will cover. Lastly the document shares details on the tool and technologies involved along with the work distribution among team members. IlmPath is a revolutionary platform that provides users the ability to learn the Holy Quran in detail with correct pronunciation. This allows users of all age and all ethnicity to correctly read the Holy Quran. The application extends to help users with any ambiguity they have in the religious through a chat box or by contacting an experienced Scholar of their choice.

1.1 Problem Statement

Indeed there is no shortage of Quran apps in the App Store or Playstore. However, many are outdated and outsized, with a poor user interface, invasive and even inappropriate ads, and unethical methods of privacy protection. We are entering a new generation of apps and are rising to the opportunity to fill a need to create a quality app that does not compromise in its purpose or execution. Its aim is to facilitate the next generation of the ummah in Learning and seeking answers in a modern, digital world. It integrates technology with religious education to provide an immersive and comprehensive learning experience for users. The platform will also serve as a valuable resource for millions of Muslims around the world who seek to improve their Quranic knowledge but have limited access to qualified educators.

1.2 Scope

IlmPath is a platform where users can learn how to read the Holy Quran with tajweed. Also, they can get more insights and knowledge about the Holy book. Plus, users can interact with the ulama and Scholars to get knowledge on any specific topic. Now diving deeper into the core features of our app. Starting from the first, users can learn how to read the Holy Quran with tajweed step by step. The app also has a Quranic chatbot that provides answers to queries based on the Quran. Moreover, users can recite any verse, and it will identify its respective Surah and Ayah number. Our next module is a platform where users can get connected to the ulama and scholars and get paid teachings from them. The application also allows users to bookmark any verse and take notes. Users can also track their learning performance to better analyze themselves. So these are all the main functionalities of the project.

Now let's discuss the boundaries and range of the proposed solution. IlmPath is mainly focused on the learning of the Holy Quran and basic knowledge of it. Plus, it provides a platform where users can get teachings from Ulama and Scholars. IlmPath does not cover the whole teachings of Islam. It does not provide anything with respect to Hadith or any other source of knowledge. It is mainly focused on learning the Quran and connecting users with scholars for any specific topic.

1.3 Modules

Following are the modules that will be integrated into the application.

1.3.1 Quranic Learning and Recitation Module

In this module, the user will recite verses of the Holy Quran, and upon making a mistake, the user will be given real-time audio feedback on the exact word they mispronounced. The user will also be given an option to start learning the Quran from basics, in which the user will be presented with Qaidah, and upon reading the Qaidah correctly, they will be allowed to move forward to the Holy Quran.

1. Qirat test
2. Learning Quran from Qaidah
3. Progress tracking
4. Root meaning of a word

1.3.2 Quranic Interaction and Search Module

Quranic chatbot will provide answers to the users' queries according to the teachings of the Holy Quran. Surah and Ayah search will work when users recite some part of the Quran. The last feature will be the recommendation of similar Ayah with it.

1. Quranic Chatbox
2. Surah or Ayah Finder Model

1.3.3 User Engagement and Customization Module

This module is for personal interaction with the Quran through noting, bookmarking Surah/Ayah, and tagging insights. It also encourages users to ponder on daily Ayahs and include personal experiences, which enable more profound and personalized learning.

1. Notes on a Surah
2. Bookmarking Surah

1.3.4 Ullama-Student Interaction Module

The application will register Ullama after a validation process that verifies the person's capability and expertise. Users can make appointments for a given time and payment with a Ullama of their own choice. The conversations will stay between a single user and their selected Ullama.

1. Ullama Registration and Verification
2. Ullama and User chat system
3. Ullama Payment based appointments

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Student/Learner	A Student is a primary user responsible for learning Quranic lessons such as starting from Qaidah and checking pronunciation. Student will be able to find a surah or ayah on reciting a part of it. Student can search for surahs on a given topic after entering their query. Moreover Students can make notes on a given surah or bookmark them. Lastly a student will make appointments with an ullama to sort any confusions they might have with a one on one conversation.
Ullama/Scholar	Scholars are individuals who are authorized users that offer Quranic teachings to the learners students. They are capable of answering inquiries, giving individual advice and recommending other paid services like personal chats or phone calls. They still, however, keep in touch with the learners under controlled circumstances.
Payment Service Provider	Payment service provider(PSP) is not a primary actor of application but it acts as the "external system actors" because they interact with the app to fulfill specific tasks like providing financial services (e.g. Process Payment, Refund Payment).
Guest	Guests are individuals that access the application to use features that don't store data in our database. These could include features such as Qirat test, Quranic chatbox and Surah finder. It was essential to add this kind of user for people who don't want to go through the hassle of registering themselves in our system to use such features.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project. Firstly lets dive into the use cases that will define our project. The use cases encapsulate and cover all the aspects of our application and defines each feature accurately.

2.1 Use-case

2.1.1 Module 1 - Quranic Learning and Recitation

USE CASE ID: 001

Name: Take Quranic Lessons

Primary Actor: Student/User

Description: Users will be able to learn Holy Quran from scratch. The users will be displayed the Qaidah and after completing the Qaidah, the user will be send to the next lesson.

USE CASE ID: 002

Name: View Lessons Progress

Primary Actor: Student/User

Description: User can view how much they have progressed in Quranic Lessons. This allows users to reflect on their performance and at how much they have accomplished with time.

USE CASE ID: 003

Name: Start Qirat Test

Primary Actor: Student/User

Description: The user initiates a voice-based test by reciting a specific ayah or surah into the app. The app listens and begins analyzing the user recitation for accuracy. The app detects any errors and presents real-time feedback via audio playback of the correct verse.

USE CASE ID: 004

Name: Recite Quran Daily

Primary Actor: Student/User

Description: The user will have the functionality to recite the Quran daily progressively. The user can continue from where they left off the next day or the next time they use the application.

USE CASE ID: 005

Name: Listen Quran Daily

Primary Actor: Student/User

Description: The user will be provided with a minute long audio of the Quran in sequence. After a minute of the audio has been played it will automatically stop and if the user does not wish to continue listening, its progress will be saved to continue from the next day.

USE CASE ID: 006

Name: Schedule Reminder for Daily Recitation

Primary Actor: Student/User

Description: Users can set reminders for daily recitation and for daily listening of the Holy Quran. A notification will be sent inviting the user to play and continue the Quranic audio.

USE CASE ID: 007

Name: View Verses Heard Progress

Primary Actor: Student/User

Description: The user will be able to view the progress of how much of the Quran he has listened to since the initial. User will be able to access how much of the audio is remaining.

USE CASE ID: 008**Name:** View Root Meaning**Primary Actor:** Student/User**Description:** The user can view the root meaning of words from an Ayah by long pressing on the word. It will provide conceptual understanding of the verse.**2.1.2 Module 2 - Quranic Interaction and Search****USE CASE ID: 009****Name:** Use Quranic Chatbot**Primary Actor:** Student/User**Description:** The student initiates a conversation with the Quranic chatbot. The system verifies user intent and prompts the student to input their query related to Quranic content. Upon entering a query, the system will represent the user with surahs/ayahs that can answer the query. The user will be able to have further interaction adding on the query.**USE CASE ID: 010****Name:** Find Surah/Ayah by Voice**Primary Actor:** Student/User**Description:** The user searches for a specific Surah or Ayah by reciting some verses or words from them. The system provides a list of surahs that contain those verses or words.**USE CASE ID: 011****Name:** View suggested surah**Primary Actor:** Student/User**Description:** The system provides the user with suggestive options as to the surah/ayah they searched for. The user will then view the suggestions and will be displayed with a list of suggestive surahs.**2.1.3 Module 3 - User Engagement and Customization****USE CASE ID: 012****Name:** Add Notes to Surah**Primary Actor:** Student/User**Description:** The user can select any ayah and add personalized notes for reflection, practice, or memorization. These notes can be saved for future reference.

USE CASE ID: 013

Name: Bookmark Surah

Primary Actor: Student/User

Description: The user can bookmark any ayah for easy access in future study sessions. The bookmark allows them to revisit important sections of the Quran efficiently.

USE CASE ID: 014

Name: View bookmarked Surahs

Primary Actor: Student/User

Description: The user will be provided with an option to view all the surahs they have bookmarked throughout their interaction with the application. The user will be able to remove or view any surah in the bookmarked list.

2.1.4 Module 4 - Ullama-Student Interaction

USE CASE ID: 015

Name: Register Scholar

Primary Actor: Scholar

Description: Researchers can create an account on the site and wait for the confirmation from the administrator. They would have their background check based upon certifications, experience and School of thought.

USE CASE ID: 016

Name: Book Appointment with scholar

Primary Actor: Student/User

Description: The user will choose a scholar from a list of all available scholars and book an appointment with them. The user will choose a time from the ullama's time slot and the respective ullama will confirm the appointment if they choose to.

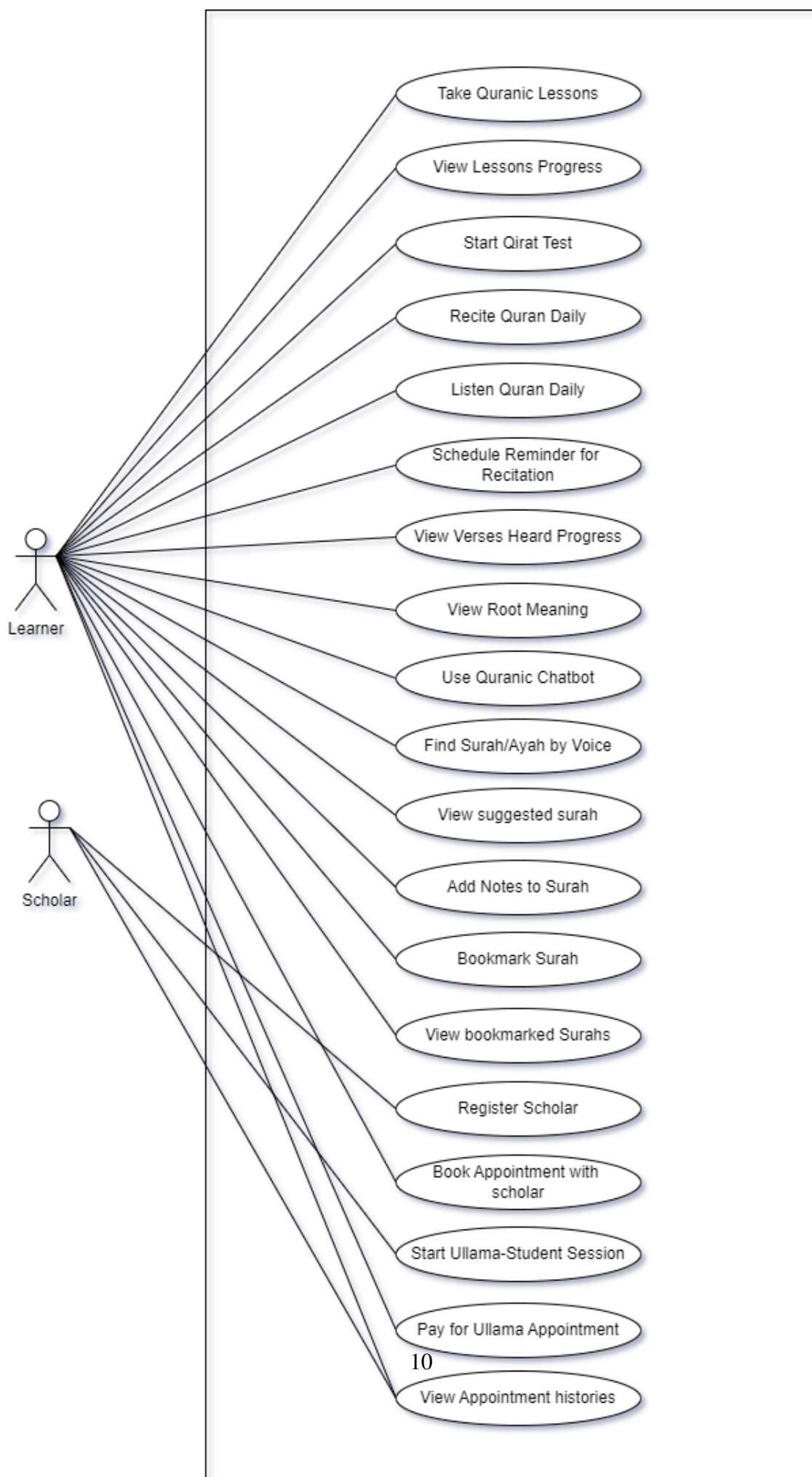
USE CASE ID: 017

Name: Start Ullama-Student Session

Primary Actor: Scholar

Description: The scholars resolve the questions among users and offer individual guidance based on the teachings of Quran and Hadith. By providing a deeper understanding of the Quranic verses, users will feel more connected and engaged with the teachings of Islam.

USE CASE ID: 018**Name:** Pay for Ullama Appointment**Primary Actor:** Student/User**Description:** Users will pay the scholar a pre-decided amount for their time and effort in solving their confusions and ambiguity.**USE CASE ID: 019****Name:** View Appointment histories**Primary Actor:** Student/User**Description:** User either student or scholars can view their appointment history. This allows users to remember and identify which scholar they had previously seek help from.



2.2 Functional Requirements

This section expresses the functional requirements derived from the use cases expressed earlier. The functional requirements give further light to the accurate functionalities of our modules and the features within.

2.2.1 Module 1

Following are the requirements for Quranic Learning and Recitation Module:

1. The system shall display the Qaidah as the initial learning resource for the student.
2. The system shall automatically promote the user to the next lesson after successful completion.
3. The system shall allow the user to browse or search for specific Surahs/Ayahs before starting a recitation test.
4. The system shall prepare for voice input once the Surah/Ayah selection is made.
5. The system shall allow users to initiate a voice-based recitation test upon request.
6. The system shall record the user's recitation and begin analysis for errors in pronunciation.
7. The system shall detect pronunciation errors or Tajweed mistakes in the user's recitation.
8. The system shall mark any detected mistakes for the purpose of live feedback.
9. The system shall provide voice feedback based on the detected errors during the user's recitation.
10. The system shall offer voice assistance, suggesting correct pronunciation through recorded audio.
11. The system shall allow students to practice Quranic recitation on a daily basis.
12. The system shall enable users to listen to the recitation while reading along.
13. The system shall display the text of the selected Ayah along with its translation.
14. The system shall provide background information and context related to the Ayah.
15. The system shall track how much time the user spends listening to Quranic recitation daily.

16. The system shall provide daily recitation progress tracking for users.
17. The system shall allow users to long-press an Arabic word to search its root meaning.
18. The system shall identify the selected word and retrieve relevant linguistic data.
19. The system shall display the root meaning and a detailed definition of the selected word.
20. The system shall provide context for both Arabic words and complex English terms.

2.2.2 Module 2

Following are the requirements for Quranic Interaction and Search Module:

1. The system shall verify the user's intent and prompt for Quranic-related queries.
2. The system shall log the conversation and ensure that appropriate Quranic references are given.
3. The system shall retrieve relevant Quranic information based on the user's query.
4. The system shall use a database of Quranic texts and Tafsir for accurate responses.
5. The system shall provide an accurate response to the user's query with Quranic references.
6. The system shall offer follow-up questions or clarification options to the user.
7. The system shall allow users to input Surah or ayah verse by recording for search.
8. The system shall display relevant search results based on the input query, including matching Surahs and Ayahs.

2.2.3 Module 3

Following are the requirements for User Engagement and Customization Module:

1. The system shall allow users to add personalized notes to specific Ayahs.
2. The system shall save the user's notes for future reference.
3. The system shall allow users to bookmark specific Ayahs for easy access.
4. The system shall display bookmarked Ayahs in the user's study dashboard.

2.2.4 Module 4

Following are the requirements for Ullama-Student Interaction Module:

1. The system shall provide users the option to book paid appointments with registered scholars.
2. The system shall offer chat option for connecting with scholars.
3. The system shall allow scholars to create a profile and submit certification documents for verification.
4. The system shall approve or reject the registration based on background checks.
5. The system shall allow scholars to respond to user queries privately with Quranic guidance.
6. The system shall notify users when a scholar has responded to their question.
7. The system shall allow scholars to set time slots for paid consultations with users.
8. The system shall process payments through secure transactions before confirming appointments.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. Numerous non functional requirements arise from the functionality of our application and some of the major non functional requirements are as follows:

2.3.1 Reliability

One of the possible failures can be seen with real time processing for Qirat test due to crashes or hangs during the process. A MTBF of 500 can be observed if the application runs for 1000 hours and encounters 2 failures after those hours. The consequence of such failure could be seen with user losing trust in the functionality of the application and can possibly result in losing the end-user. To avoid such failures to occur, vigorous testing in the form of unit testing and blackbox testing can be observed.

2.3.2 Usability

Usability requirements deal with ease of learning, ease of use, error avoidance and recovery, the efficiency of interactions, and accessibility. The usability requirements specified here will help the user interface designer create the optimum user experience. The following are the non functional requirements that incorporates this point:

USE-1: The user interface should be simple and intuitive, allowing users of all ages to interact with the system without requiring technical knowledge.

USE-2: The app should provide clear audio feedback and easy-to-navigate text for recitations, ensuring clarity in teaching and assessments.

2.3.3 Performance

To ensure the application runs smoothly, we need to implement some performance checks and measures. The following are the non functional requirements that incorporates this point:

PER-1: The system shall be able to provide real time feedback in the features like voice-based Quranic feedback and chatbot, with very less response time. The response time for the system to find the results should not exceed 10 seconds of the user input.

PER-2: System must be able to manage many users at the same time. The system must be able to make payment transactions smooth without any error. The system shall ensure fast data retrieval for user progress tracking and performance analytics, with updates to user records being processed in under 5 seconds.

2.3.4 Security

Our application ensures security of user data. This provides users assures users to use the application without fear of exposing their data, The following are the non functional requirements that incorporates this point:

1. User Authentication and Authorization

Only logged-in and authorized individuals shall be permitted access to notes and bookmarking features. This is to safeguard users' private information from being accessed or altered by intruders, who are not the rightful owners of the information.

2. Data Privacy and Integrity

The software has to guarantee that meanings and notes/bookmarks from the user are not altered by any outside party. A low level of security will make it easy, for example, to introduce security check that ensure no changes to user notes occur within an application without having implemented any authorisation concepts. The system will also ensure complete transparency for chats, as no user or ullama can see other users or ullama chats.

2.4 Domain Model

The following is our representation of the project through a domain model. The classes and their relations accurately describe how our project works.

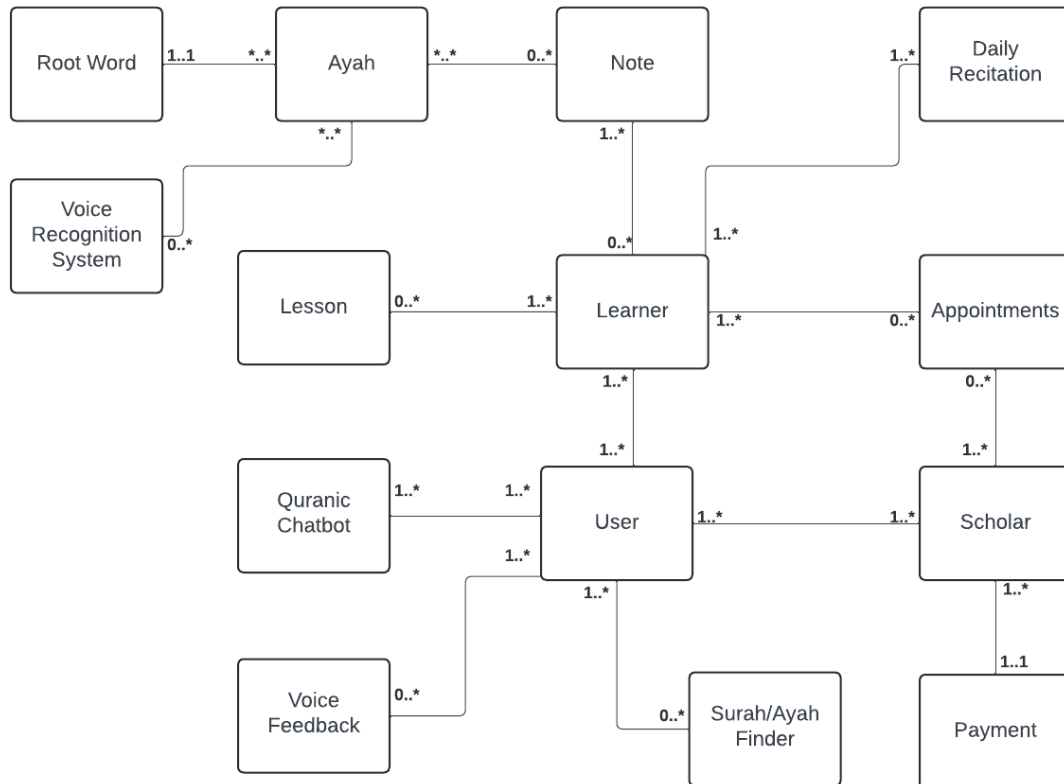


Figure 2.2: Domain Model

Chapter 3

System Overview

IImPath aims to provide an experience that allows users to correct their pronunciation, search queries and get answers in the light of the Quran, find Surahs and tackle any ambiguity they have with the guidance of ullamas. This section provides an overview of many designs that help understand this project and it's functionalities. The section covers illustrations of Architectural design, Activity Diagram, Class Diagram, Class-Level Sequence Diagram, State Transition Diagrams and an ERD to give representation of all the functionalites and their workflow.

3.1 Architectural Design

Our project can best be represented in a Multi-Tiered Architecture that composes of three layers. The first layer is the presentation layer that showcases the user interface i.e our React Native application. The second layer contains all the business logic that will be used to develop the application. This contains lessons and qirat testing, Quranic chatbot, Surah finder, daily recitation and scholars platform, all part of the business logic. The third and final layer encapsulates the data layer. This represents the models that process data, database for storage and retrieval of data and payment gateways that ensure accurate transactions.

Multi-Tiered Architecture

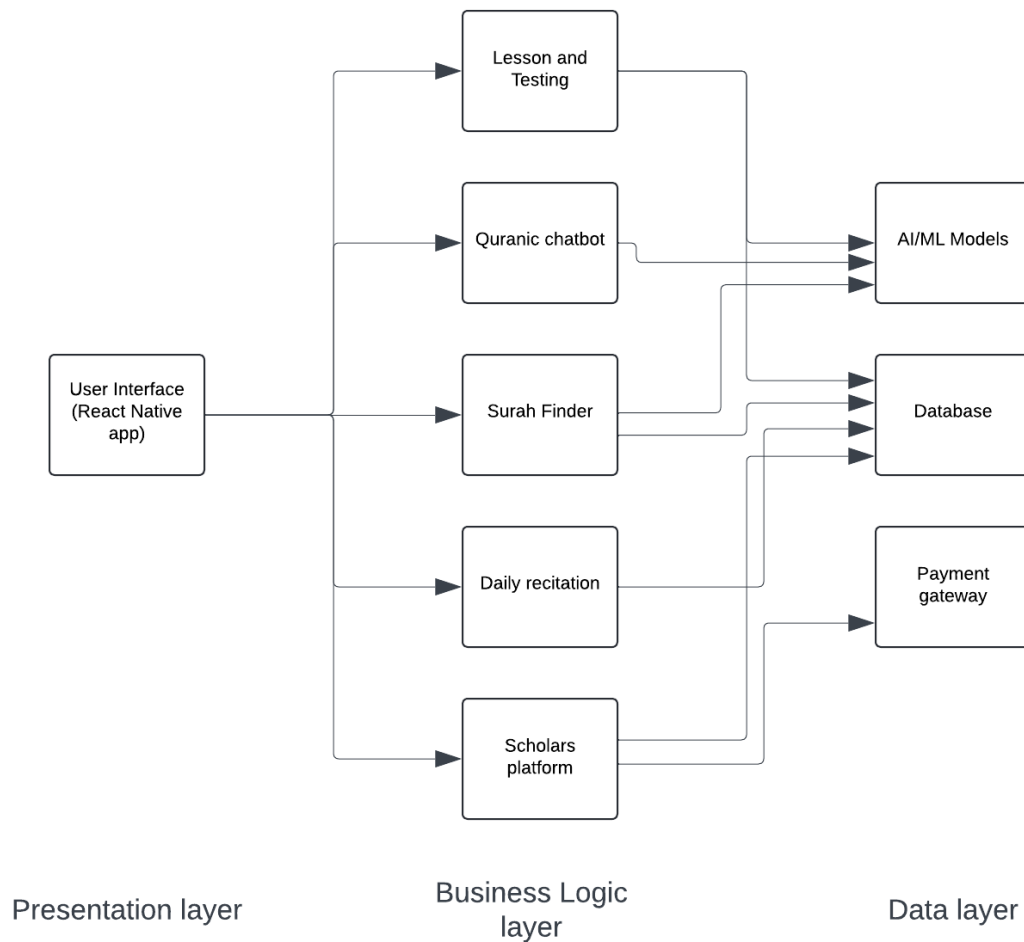


Figure 3.1: Architectural Diagram

3.2 Design Models

The design models selected use the Object Oriented Development Approach. The design models comprise of illustrations of activity diagram, class diagram, state transition diagram and class-level sequence diagram that accurately describes the development approach and workflow of the application.

Design Models for Object Oriented Development Approach

• Activity Diagram

• Class Diagram

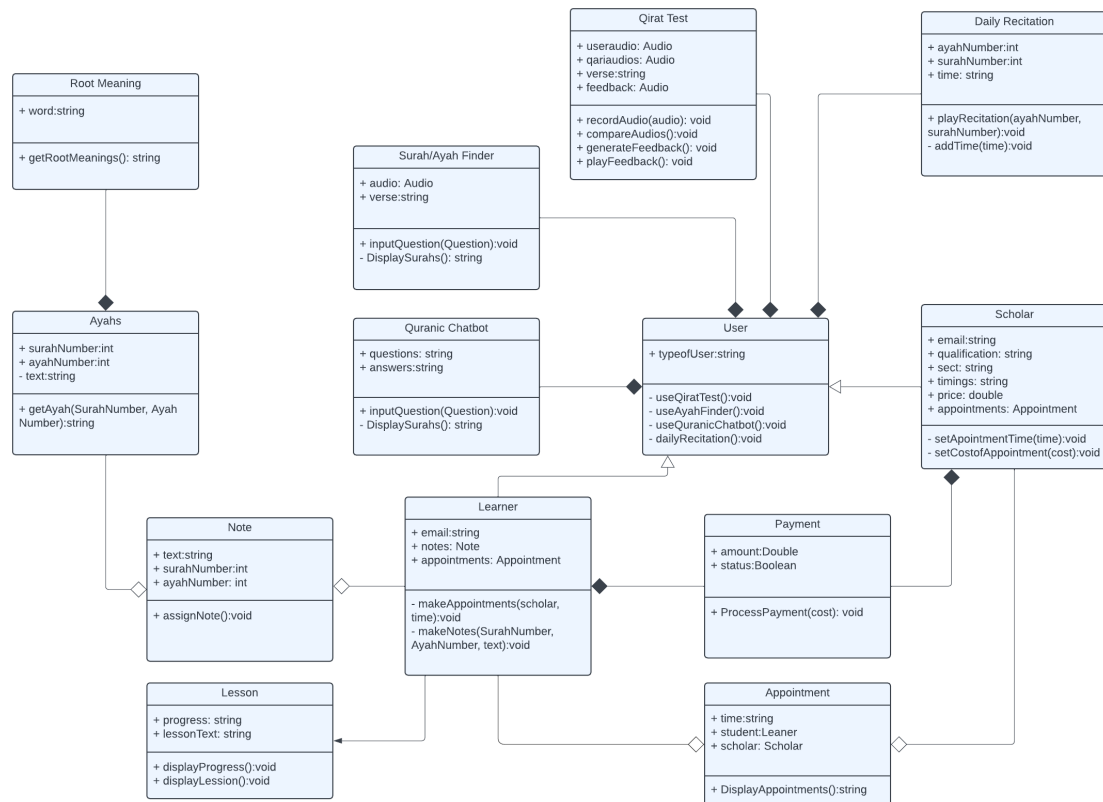


Figure 3.3: Class Diagram

• Class-level Sequence Diagram

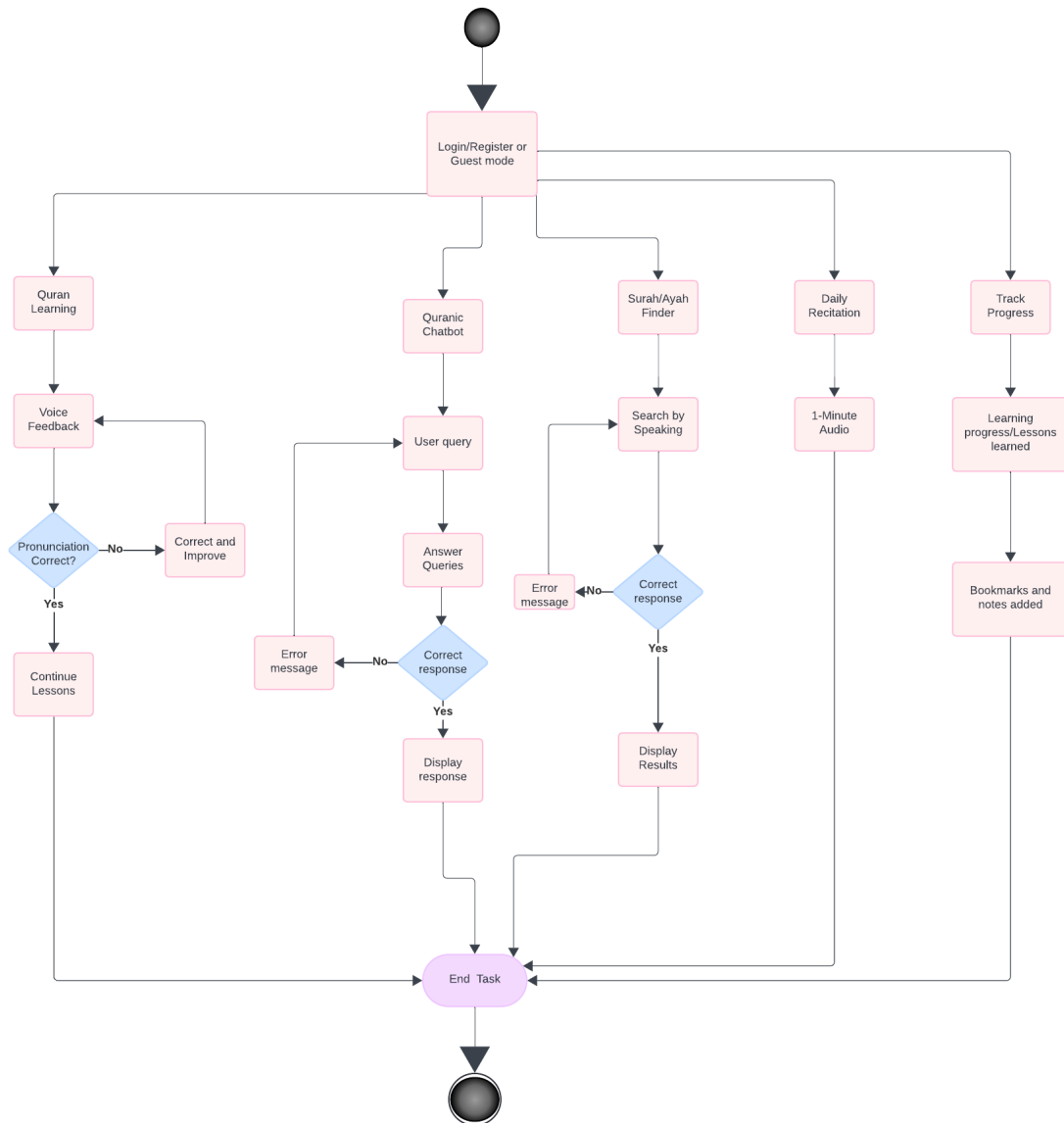


Figure 3.2: Activity Diagram

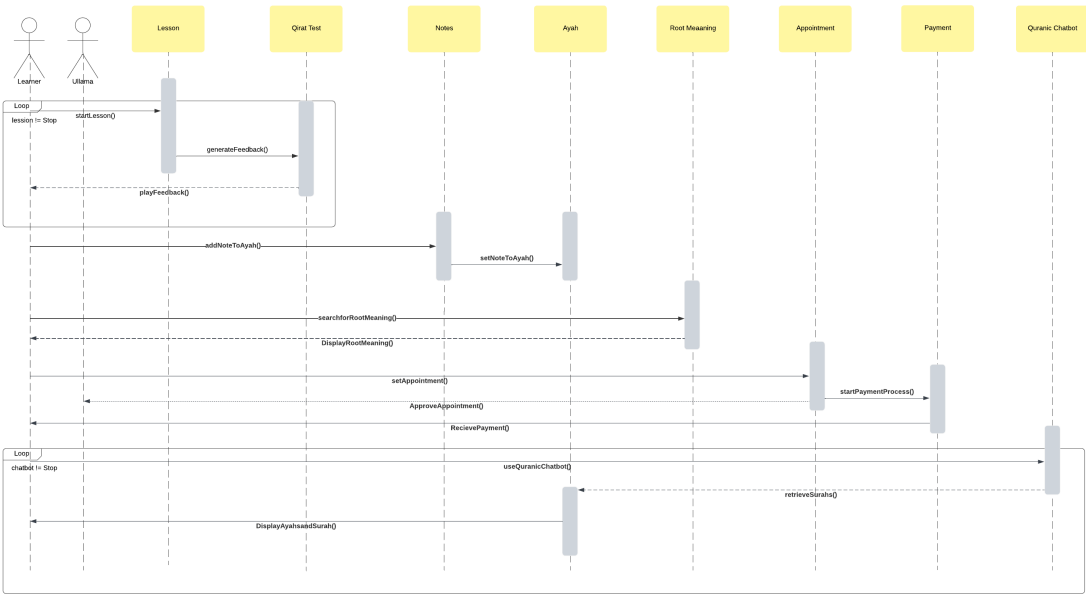


Figure 3.4: Class-Level Sequence Diagram

• State Transition Diagram 1

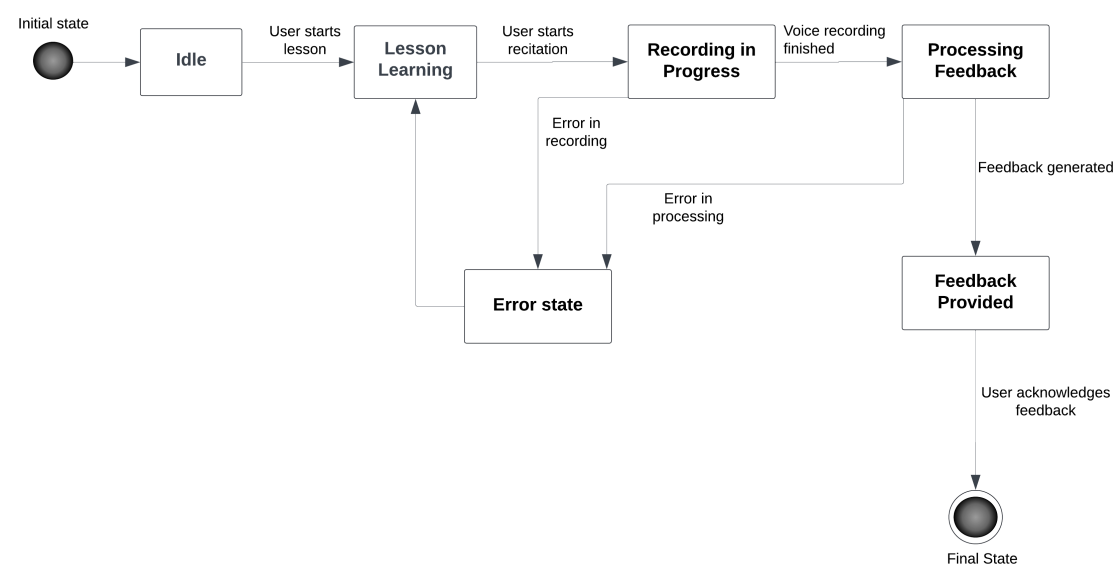


Figure 3.5: State Transition Diagram 1

• State Transition Diagram 2

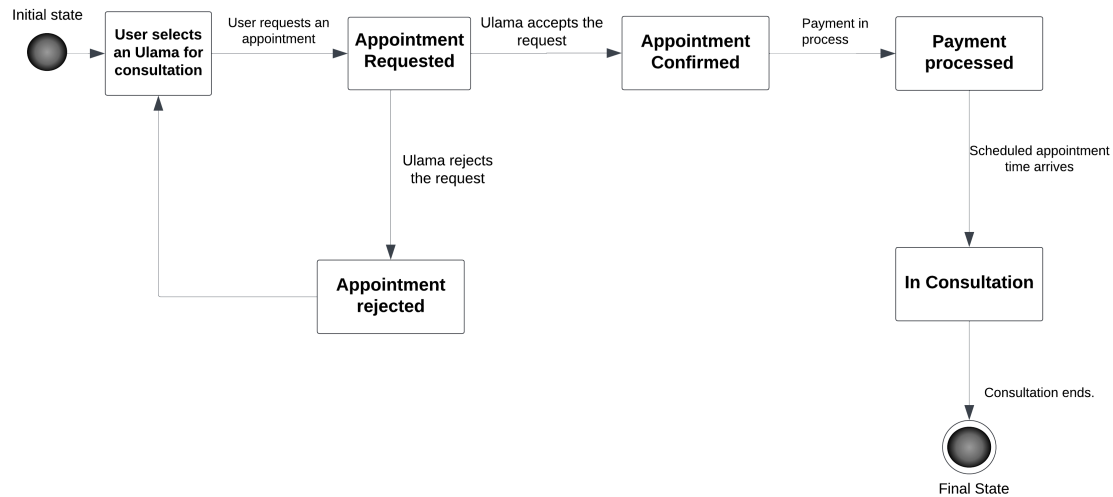


Figure 3.6: State Transition Diagram 2

• State Transition Diagram 3

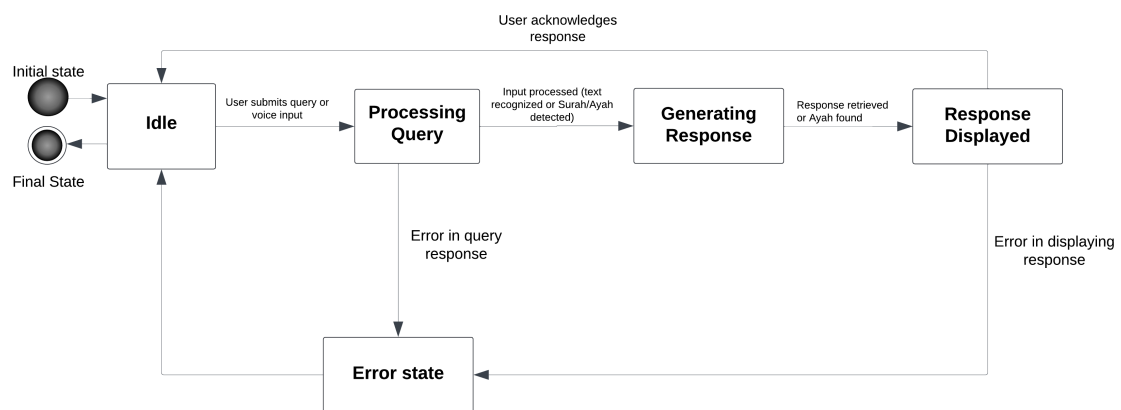


Figure 3.7: State Transition Diagram 3

3.3 Data Design

Many features in our application require the use of a database. The database used will comprise of tables that ensure storage and retrieval of data. Firstly any user registered, either learner or scholar will have their data stored in the database. Next any notes or bookmarks a user makes will get stored in the database so that they can view them later. Similarly to maintain record of appointments or to track progress of users, database is

essential. The following Entity Relationship Diagram represents it accurately.

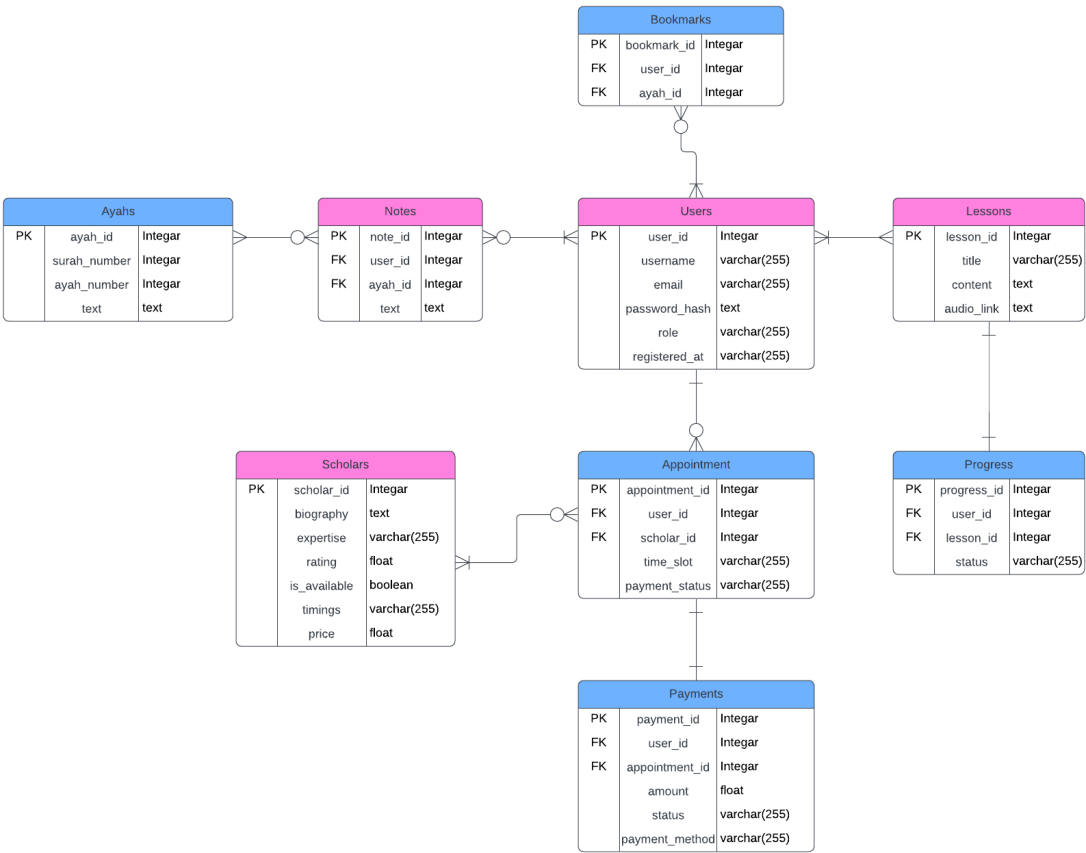


Figure 3.8: Entity Relationship Diagram

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